Durva Devendra Kumar

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Scholastic Achievements

- Received a grade of **AA/AB** (9+) in **7** diverse core courses showcasing continued academic excellence *(2024-2026)*
- Ranked within the top **1.3%** out of **1.2M+** candidates appearing for the JEE Mains Examination *(2024)*
- Maintained **top 3** ranking from **elementary** to **high school**, demonstrating unwavering academic dedication *(2014-2021)*

Research Experiences

STAR Lab | *Undergraduate Researcher* | *Mechanical subsystem* *(Aug'26 - Present)*

- Designing the mechanical architecture of a **customised, launch-compatible star tracker**, developing the structural and mounting components while considering packaging, alignment, and manufacturability constraints

Identifying GRB signals | *KSP 6.0* | *Guide: Dr. Gaurav Waratkar, Suryansh Srijan* *(May'25 - Jul'25)*

- Processed **AstroSat CZTI** and **Veto detector** event-mode FITS data for **gamma-ray transient** detection analysis
- Applied time-binning, SAA masking, smoothing, and detrending across **multiple bin sizes** to prepare the datasets
- Implemented an **N-sigma** peak detection algorithm, applying multi-quadrant coincidence checks to analyze **19 candidate transients** through the energy bands **20–500 keV** across all CZTI quadrants, confirming **1 GRB event**
- Generated and interpreted **spectrogram grids** as the final stage of the pipeline, validating GRB spectral signatures

Imaging a Bright Neutron Star | *KSP 6.0* | *Guide: Kenil Ajudiya, Ayushi Chippa* *(May'25 - Aug'25)*

- Researched the fundamentals of **interferometry** to understand how multiple antennas combine their signals coherently
- Utilized **CASA** to process a **GMRT dataset** containing **88 minutes** of observations through the imaging pipeline
- Performed **10+ rounds** of iterative **flagging** and **calibration**, across **256 different channels** (2.0 MHz each)
- Generated a preliminary dirty image of the supernova remnant **B0329+54** using **self-calibration**, and applied **deconvolution** with the CLEAN algorithm **(1200+ iterations)** to reconstruct a restored image of the source

Detection of Solar Flares | *K-CAP* | *Guide: Ashwin Goyal* *(Dec'24 - Jan'25)*

- Processed over **1 month** of **Aditya L1** solar data, defining a method to identify the position and duration of flares
- Developed and applied a **Gaussian rise + exponential decay** curve fit to characterize flares and enable comparison
- Achieved an average R^2 of **0.81** and a maximum R^2 of **0.94**, demonstrating **strong predictive performance**
- Reduced false positives through **iterative background modeling**, improving robustness of detection across datasets

Technical Experiences

IIT Bombay Racing Team | *Mechanical JDE* | *Drivetrain subsystem* *(Feb'25 - Jul'26)*

Faculty Advisor: Prof. Parthib Khoud, Prof. Nived Chebrolu and Prof. Archak Mittal
A 3-tier cross-functional team comprising 70+ students who design and fabricate a Formula-style electric race car

Formula Student 2026 | *Formula Student Switzerland* | *Formula Student Portugal* *(Jul'26)*

- Part of the **40-member contingent** representing IIT Bombay Racing at **FS Switzerland** and **FS Portugal**
- Achieved **Overall Champions** in the **EV and DV category** at **Formula Student Portugal 2026**, with **1st** in EDP, and **2nd** in Cost Event, following a **13th-place** overall finish at **Formula Student Switzerland 2026**
- Prepared **Costed Carbonized Bill of Materials** for the Engine and tractive subsystem, providing lifecycle data including manufacturing cost breakdown and quantified carbon emissions for **Cost and Manufacturing event**

Formula Bharat 2026 | *IIT Bombay Racing Team* *(Jan'26)*

- Achieved **3rd Overall (Statics + Dynamics)** at Formula Bharat 2026, along with 1st Place - Best Powertrain System Award, 1st Place – Cost Event, 3rd Place – EDP and **1st Place – Overall Statics** at national level

Junior Design Engineer | *IIT Bombay Racing Team* *(Feb'25 - Jul'26)*

- Assisted in the assembly and maintenance of the **drivetrain and wheel assembly** of the Formula student car
- Showcased implemented **drivetrain design** changes to achieve performance improvements during the Formula Student **Design Presentation**, emphasizing gearbox optimization, motor selection, and cooling system design
- Contributed to the selection of the final **compound planetary gearbox configuration** for the next-generation vehicle, accounting for **size and weight constraints** while optimizing the drivetrain architecture
- Developed a **turbulent flow model** on MATLAB to evaluate coolant flow for motor cooling jacket design
- Designed the motor cooling jacket geometry using Solidworks and characterized coolant flow behaviour through **COMSOL CFD simulations**, enabling design validation and refinement before manufacturing
- Designed the motor shaft geometry using CAD and validated **structural integrity** through ANSYS simulations
- Developed a bearing selection **MATLAB tool** to automate bearing calculations, accelerating component selection

Key Projects

Deadbeat Escapement Analysis | *Course Project* *(Feb'26 - Apr'26)*

Guide: Prof. P. Seshu for ME444 Analysis and Design of Mechanical Systems

- Studied the **escapement mechanism** through kinematic and dynamic analysis, and fabricated and assembled a **3D-printed mechanical clock prototype** to validate the analytical findings experimentally
- Performed pendulum calculations, selecting a **876 mm pendulum length** to achieve the required time period
- Implemented a **Python-based simulation** to analyse **pendulum motion** and **escapement-driven oscillations**
- Evaluated performance limitations through **friction analysis** and **0.01 mm tolerance effects** on mechanism working

RUL Prediction of Turbofan Engines | *Course Project* *(Feb'26 - Apr'26)*

Guide: Prof. R. Ganesh and Prof. Neeraj Kumbhakarna for ME228 Applied Data Science and Machine Learning

- Developed **Remaining Useful Life** (RUL) prediction models for turbofan engines using **4 NASA C-MAPSS** datasets
- Reduced 21 sensors to 14 via **Pearson correlation**, achieved $R^2 = 0.850$ on FD004 using a **BiLSTM model**
- Identified implicit **fault modes** in FD003/FD004 datasets using **K-Means clustering** on engine degradation features and validated distinct degradation patterns through **PCA** and **sensor trajectory analysis**
- Achieved **25% reduction** in late predictions: **BiLSTM** 5,060 outperforming **LSTM** 6,812 on **NASA score**

Stress Analysis of Perforated Plates | *Course Project* *(Mar'26 - Apr'26)*

Guide: Prof. Sripriya R. for ME218 Solid Mechanics Lab

- Analysed stress distribution using **photoelasticity and FEA simulations** across **3 geometric configurations**
- Fabricated and tested epoxy specimens with **3D-printed TPU moulds** under controlled loading conditions
- Implemented **ANSYS simulations** to compare stress contours and experimental fringe patterns at **20N loading** and evaluated the stress concentration factors for fillets and holes achieving **SCF values** of **1.14–1.7 across geometries**

RC Plane | *Aeromodelling Club | IIT Bombay* *(Sep'24 - Nov'24)*

- Designed a **middle-wing RC trainer** aircraft using **Depron foam** after analyzing multiple wing configurations
- Integrated electronic components, including a **Brushless DC motor**, **Li-ion battery**, and **servo motors** to successfully develop a reliable and efficient **actuation system** that delivers precise and consistent motion control
- Achieved a successful flight demonstration with **controlled take-off**, **stable in-flight maneuvers**, and **safe landing**

Drone controller | *Course Project* *(Aug'24 - Nov'24)*

Guide: Prof. Joseph John for MS101 Makerspace

- Collaborated in a team of **6 students** to design and build a drone controller, using **3D modeling** for custom fabrication
- Developed a custom PCB using **EasyEDA** to integrate a microcontroller, dual joysticks, and communication modules
- Programmed the controller to interpret **dual analog joystick inputs** and transmit commands wirelessly to the drone

XLR8 | *Electronics and Robotics Club | IIT Bombay* *(Aug'24 - Sep'24)*

- Engineered a robotic car in a team of 4 by programming a **Raspberry Pi Pico W** to control motors via an **L298N** driver
- Designed and soldered a custom perfboard controller integrating **MPU6050** and **ESP01** modules for gesture detection
- Built a **sensor-equipped remote** for maneuverability and managed the Li-ion battery system for efficient power use

Positions of Responsibility

Department Academic Mentor | *SMP Team | IIT Bombay* *(Jun'26-Present)*

*Selected in a **52-member team** from 120 applicants after a rigorous interview and peer-review process*

- Guiding and mentoring **6 sophomore students** in their academic, personal and extracurricular pursuits
- Contributing to the **Core Resources** subgroup, developing resources and providing guidance to prospective students

Convenor | *Krittika Club | IIT Bombay* *(Apr'25 - Apr'26)*

Worked in a 14-member core team, organizing club events and managing logistics for the smooth execution of activities

- Conducted **Telescope Learning Sessions** for 6 students, teaching observational techniques and telescope operation
- Organised a high-profile exhibition of different astrophotographs for **National Space Day** at IIT Bombay, in collaboration with **The Sky Explorers**, engaging **13k+ students** and **100+ residents** with an interest in astronomy
- Conducted sessions for **Theory Learners' Space**, a summer club course with **10+ modules** on theoretical astronomy, covering topics such as **radio astronomy**, **solar physics**, and **photometry and spectroscopy**

Technical Skills

Programming	C++, Python, MATLAB, LaTeX, G-code
Libraries	Numpy, Pandas, Matplotlib, PyTorch, SciKit, Tensorflow, Seaborne, Plotly
Software	Jupyter, Fusion360, SolidWorks, Generative Design, COMSOL Multiphysics, MATLAB, ANSYS (Static Structural), Simulink, Simscale

Key Courses Undertaken

Mechanical Core	Heat Transfer* | Control Systems* | Fluid Mechanics | Mechanical Processing of Materials | Analysis and Design of Mechanical Systems | Applied Thermodynamics* | Structural Materials | Thermodynamics | Solid Mechanics | Kinematics and Dynamics of Machines | Thermal and Chemical processing of materials*
Computer Science	Applied Data Science and Machine Learning | Computer Programming (C++)
Mathematics	Calculus I | Calculus II | Differential Equations | Linear Algebra
Other Courses	Finite Element and Boundary Element Methods* | Introduction to Aerospace Engineering | Economics | Psychology | Entrepreneurship | Biology

Extracurricular Activities

**to be completed by Nov '26*

- Taught under **Voice for Purpose** (**NSS IIT Bombay**), supporting learning for the **visually impaired** *(2024)*
- Completed an online **Python** training programme with AI, covering fundamentals, database connectivity, and applications across various disciplines, culminating in a project on building a **Fantasy Cricket League game** *(2021)*
- Completed a **Udemy** course in **Deep Learning with Python**, covering Python and neural network concepts *(2021)*
- Participated in a full-day workshop on **Space Telescopes**, covering their scientific role, launch, operation, and maintenance, with hands-on experience in **JWST model making** and interactive orientation sessions *(2021)*
- Served as **Discipline Incharge** in the **Junior Student Council**, ensuring adherence to conduct guidelines *(2015)*